

SensorMultiHeadDecoder V2 Architecture

Sweep V7: Window Position + File-Level MWC + Pointer Network + SensorValuePrior + DropPath · ~6.1M params

V7 NEW features highlighted in green

MM-DTAE-LSTM Encoder (frozen, ~8.6M params)

IoT sensors -> per-modality encoding -> cross-modal fusion -> DTAE -> LSTM · d_model=256, w=256, s=64 · 155 features
Weights frozen · Provides sensor embeddings [B, T_s, 128] · Operation prediction [B, 9]

Decoder Inputs

Five sources condition the autoregressive token generation (V7: +window position, +file-level neighbors)

Sensor Embeddings

From frozen encoder
Latent representations of IoT sensor time series
[B, T_s, 128]

Operation Type

100% accurate label
9 CNC operations
adaptive/face/pocket x 3
[B]

V7 Window Position

87% info gain for Y-values
window_index (0-15)
+ total_windows + frac
[B] + [B]

V7 File-Level MWC

2-4 adjacent windows
Actual file-level neighbors (fixed V5 random bug)
[B, k×T_s, 128]

G-code Token Sequence

Target tokens (teacher forcing)
BOS-prefixed IDs from 712-token vocab (0% UNK)
Train: scheduled sampling 0.3-0.5 · Infer: greedy/beam
[B, L]

Operation Conditioning + Sensor Projection

Conditions sensor memory on the CNC operation type · ~88K params

Sensor Embeddings

[B, T_s, 128]
from frozen encoder

Operation Embedding

nn.Embedding(9, 96)
[B] -> [B, 96]

Broadcast to [B, T_s, 96]

Concatenate [B, T_s, 128 + 96] = [B, T_s, 224]

Sensor Projection MLP

Linear(224->384) -> LayerNorm(384) -> GELU -> Dropout(0.1)
[B, T_s, 224] -> [B, T_s, 384]

Memory Enrichment (V7 New)

Window position, file-level multi-window context, and memory positional encoding

Window Position Features

Embed(max_pos=32, dim=96) -> pos_emb [B, 96]
Linear(1->48) -> frac_emb [B, 48]
cat + Linear(144->384) -> broadcast + add to memory

File-Level MWC Neighbors

Lookup by source_file + window_index
k = 2 or 4 neighbor windows (Cohen's d=1.09)
Concatenate neighbor memory to main memory

Memory Positional Encoding

Sinusoidal PE (max_len=4096, dropout=0.1)
Tells decoder WHERE in the toolpath
each window falls: disambiguates Y-values

memory = sensor_proj + win_pos | cat(memory, neighbor_memories) | memory_PE(memory)

memory [B, (1+k)×T_s, 384]

Token Embedding

~273K params

nn.Embedding(712, 384)
scale by sqrt(384) + Embed Dropout(0.1)

+ Sinusoidal Positional Encoding
max_seq_len = 16 · no learnable params

tgt [B, L, 384]

shift right
y_{t-1} -> input

K, V

Transformer Decoder

8 layers · Pre-LN · ~4.7M params · d_model=384 · d_ff=1536 · Gradient Checkpointing

TransformerDecoderLayerWithDropPath

x8

Masked Multi-Head Self-Attention

8 or 12 heads · d_k = 48 or 32 · Causal mask · + Residual · + DropPath

Multi-Head Cross-Attention to Sensor Memory

Q: token [B, L, 384] K, V: memory [B, (1+k)×T_s, 384] 8/12 heads · + Residual · + DropPath

Position-wise Feedforward Network

Linear(384->1536) -> GELU -> Dropout(0.1) -> Linear(1536->384) · + Residual · + DropPath

V7 DropPath / Stochastic Depth: linearly increasing 0 -> 0.1 across 8 layers (drops entire residual branches)

[B, L, 384]

LayerNorm(384)

hidden [B, L, 384]

Each: Lin(384->192) -> LN -> GELU -> Drop -> Lin(192->N) · CE loss

Type Head

Token category -> 4 classes

SPECIAL / COMMAND
PARAM / NUMERIC

[B, L, 4]

Command Head

G-code command -> 6 classes

G0 · G1 · G2 · G3
G53 · OTHER

[B, L, 6]

Param Type Head

Parameter letter -> 10 classes

X · Y · Z · F · R
S · I · J · K · OTHER

[B, L, 10]

Digit-by-Digit Value Head

op_type + param_type -> context -> sign(3) + 6 digits(10) · CE+MSE

Sign Head
[B, L, 3]

Digit Heads x6
[B, L, 6, 10]

Aux Reg
[B, L, 1]

~480K params

V7 Pointer Network (Per-Axis Classification)

Per-axis heads from hidden [B, L, 384] · ~260K params

X Axis

384->192->203
203 values

Y Axis

384->192->258
258 values

Z Axis

384->192->224
224 values

R Axis

384->192->7
7 values

Each: Lin->LN->GELU->Drop->Lin · per-axis CE · pointer_weight = 1.0 or 2.0
Directly attacks X/Y value confusion
Values from data/axis_value_tables.json

V7 Sensor Value Prior (Bypass Path)

Direct sensor -> digit path (bypasses transformer) · ~111K params

Input: raw sensor embeddings [B, T_s, 128] (from frozen encoder)

Global pool -> Lin(128->64) -> LN -> GELU -> Drop

10 per-param-type digit heads: 64->64->60 (6 digits x 10)

10 per-param-type sign heads: 64->32->3

Output: digit_bias [B, L, 6, 10] + sign_bias [B, L, 3]

Added to digit/sign logits (weight=0.5)

CE Legacy Token Head

Linear(384 -> 712)

+ Type Constraint Mask

Mask invalid tokens per type pred

+ Grammar FSM Mask

BOS->CMD/PARAM, PARAM->NUM

NUM->PARAM/EOS, CMD->CMD

[B, L, 712]

~274K params

Type-Constrained Token Assembly

The type head determines which sub-head generates each token position

Type = SPECIAL (0) -> PAD/BOS/EOS

Type = COMMAND (1) -> Cmd Head

Type = PARAM (2) -> Param Head

Type = NUMERIC (3) -> Digits + Pointer + Sensor Prior (e.g. +01.5750)

G-code Token Sequence Output

Autoregressive generation of G-code from sensor data

BOS G1 X +01.5750 Y -00.3200 Z +00.0000 F +15.0000 EOS

SPECIAL COMMAND PARAM NUMERIC (digit-by-digit + pointer)

Training Configuration (V7)

Loss: weighted sum of CE (type, cmd, param, sign, digits) + MSE (aux) + pointer CE + legacy CE

AdamW (lr=3e-4, cosine) · Focal (gamma=2.0) · Label smoothing 0.1-0.2 · Weight decay 0.05-0.1

500 epochs, patience 120 · Scheduled sampling 0.3-0.5 · Batch 32 · DropPath 0.0-0.1

Encoder frozen · Causal autoregressive · Stratified 5-fold CV (sequence coverage repaired)

loss = w_type·CE_type + w_cmd·CE_cmd + w_param·CE_param + w_digit·(CE_sign+CE_digits) + w_legacy·CE_legacy + w_ptr·CE_pointer

Parameter Summary

Conditioning: ~88K · Embedding: ~273K · WinPos: ~59K · Transformer x8: ~4.7M · Heads: ~595K · Pointer: ~260K · SensorPrior: ~111K · Total: ~6.1M trainable